이승철

A. 교수님과 연구실 소개

교수님 약력

- **한국과학기술원 기계공학과 부교수 (23′~현재)**
- 포항공과대학원 기계공학과 조·부교수 (18′~23′)
- 울산과학기술원 기계공학과 조교수 (13′~17′)
- University of Michigan, Ann Arbor 박사 (10′)
- University of Michigan, Ann Arbor 석사 (08′)
- 서울대학교 학사 (01′)

- 한국생산제조학회 제조AI 부문 위원장
- 대한기계학회 신뢰성 부문 부회장

연구실 정보

- 주소: 대전광역시 유성구 대학로 291
- 전화: 042-350-3026
- 이메일: seunglee@kaist.ac.kr
- 연구실명: 산업인공지능 연구실
- Google Scholar: *https://scholar.google.com/citations?user=AeMlKBAAAAAJ*
- 연구실 홈페이지: *https://iailab.kaist.ac.kr/*



대표 연구 키워드

AI+X: Physics
- Multiphysics Simulation AI
- Geometry-adaptive AI

AI+X: Materials
- Active Learning for Material Discovery
- AI-driven Process Optimization

AI+X: Acoustics
- Sound Source Localization
- Acoustic Scatterer Design

AI+X: Healthcare
- AI-based Medical Image Analysis
- Organoid Analysis & Diagnostics

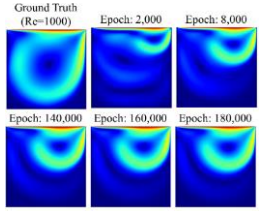
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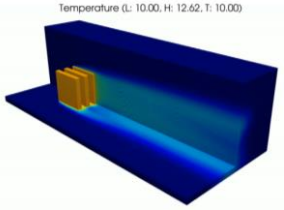
B. 주요 연구성과

AI + Physics

- Multiphysics Simulation with Neural Network
- Repetitive Parameter Initialization PINN
- Geometry-adaptive PI-DeepONet

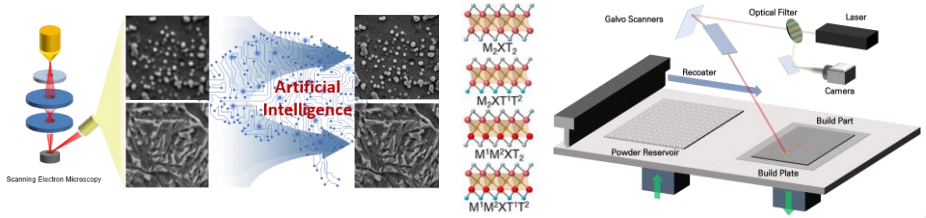






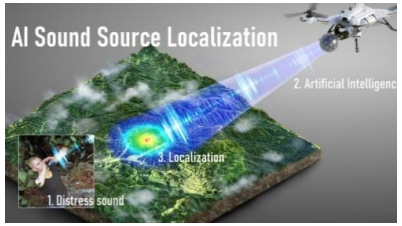
AI + Materials

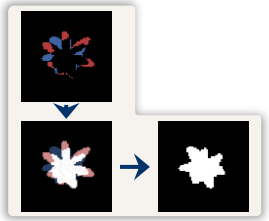
- Advanced Microscopy Processing
- High-throughput Screening for DFT/MD Calculations
- Process Optimization for Additive Manufacturing

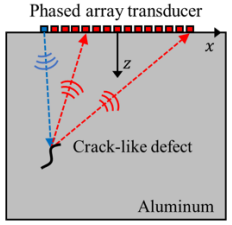


AI + Acoustics

- Sound Source Localization
- Physics-based Acoustic Scatterer Design
- Deep Learning-based Ultrasonic Image Translation

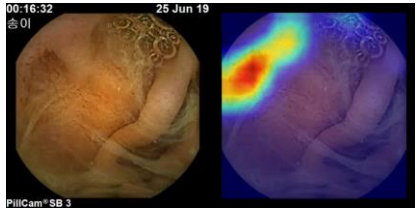


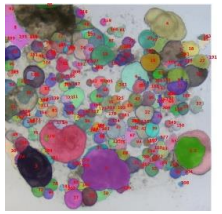


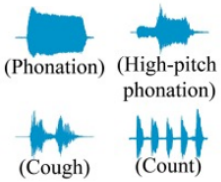


AI + Medical

- Polyp Detection for Small Bowel Capsule Endoscopy
- Deep learning-based Organoid Segmentation
- Deep Learning-based Voice Analysis





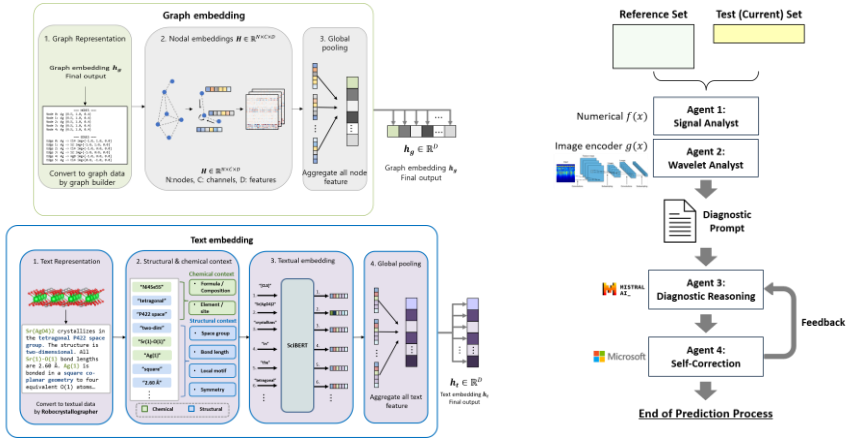


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B. 주요 연구성과

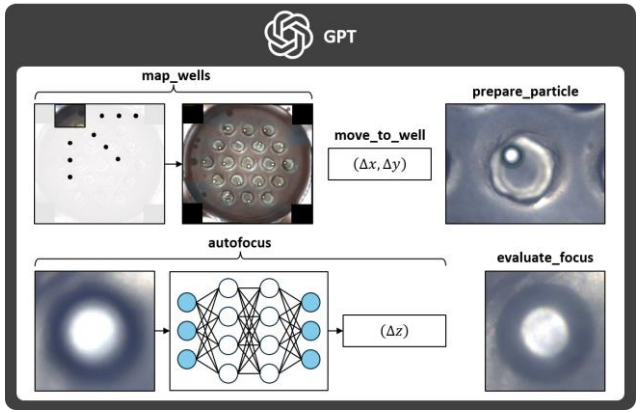
LLMs for Engineering

- LLM-based 2D Material Property Prediction
- Multi-Agent LLM for Bearing Fault Diagnosis



Physical AI

- LLM Agent-based Microscope Autofocusing



주요 연구 및 교육 실적

- 연구과제 58건 수행 (진행 8건, 완료 50건)
- 특허 등록 8건 및 출원 6건
- 기계인공지능 관련 강습회 75회
- iAI KAIST 유튜브를 통한 온라인 AI 강의 제공
- 『기계상태진단 개론』 공저

주요 수상 경력

- ’26. 산학협력 유공 교육부 장관표창
- ’25. 대한기계학회 김윤영 AI교육혁신상
- ’25. 한국소음진동공학회 학술상
- ’23. 대한기계학회 CAE 및 응용역학 젊은연구자상
- ’22. 기계의 날 산학연 협력 유공 산업통상자원부 장관표창